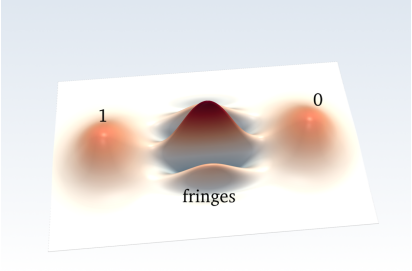
Stabilized cat qubits

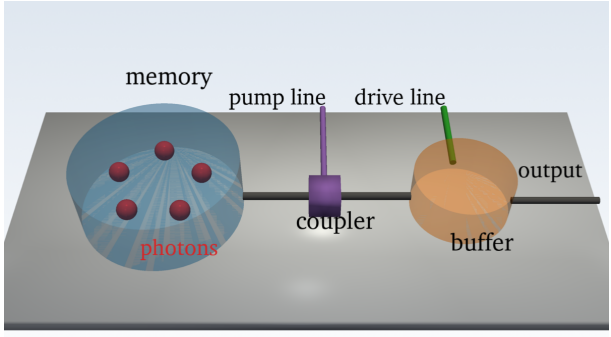
DuQuantum 2026, Alice & Bob challenge

Team Shunya



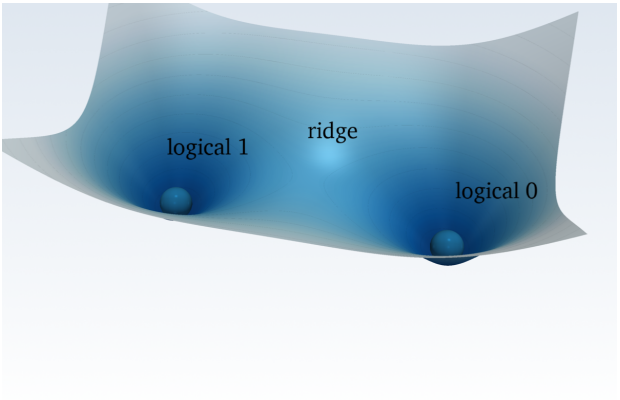
The setup

- A storage resonator holds the qubit, about 4.4 photons of microwave light
- A pump and a leaky helper resonator keep it in shape; we simulated both together



The qubit is a ball in one of two valleys

- One valley means 0, the other means 1, and the pump keeps the ball where it is
- Switched on from empty, the pump builds the qubit in about 1 μs



Bit flips are a million times rarer than phase flips

$2.6\ \mathrm{s}$	$1.94\ \mu\mathrm{s}$	10^6
time to a bit flip	time to a phase flip	how much rarer bit flips are

- A bigger cat makes the valleys deeper, so the ball almost never hops
- Only one kind of error is left to correct, which makes error correction cheaper

Reading the qubit

- The helper's output goes up for 0 and down for 1, readable within about 1 μs
- Best readout error 0.4%; enlarging the cat while reading cuts it 13 to 18 times

